

Problem Analysis Chart (PAC)

A **Problem Analysis Chart (PAC)** is a simple table used in problem solving (especially in programming) to clearly identify:

- **Input** → What data is given
- **Process** → What steps/logic to apply
- **Output** → What result is expected

Find Square of a Number

Input	Process	Output
Number n	Square = $n \times n$	Square of n

Find Simple Interest

Input	Process	Output
P, R, T	SI = $(P \times R \times T) / 100$	Simple Interest

Find Largest of Two Numbers

Input	Process	Output
a, b	If $a > b \rightarrow$ print a, else print b	Largest number

Area of Circle

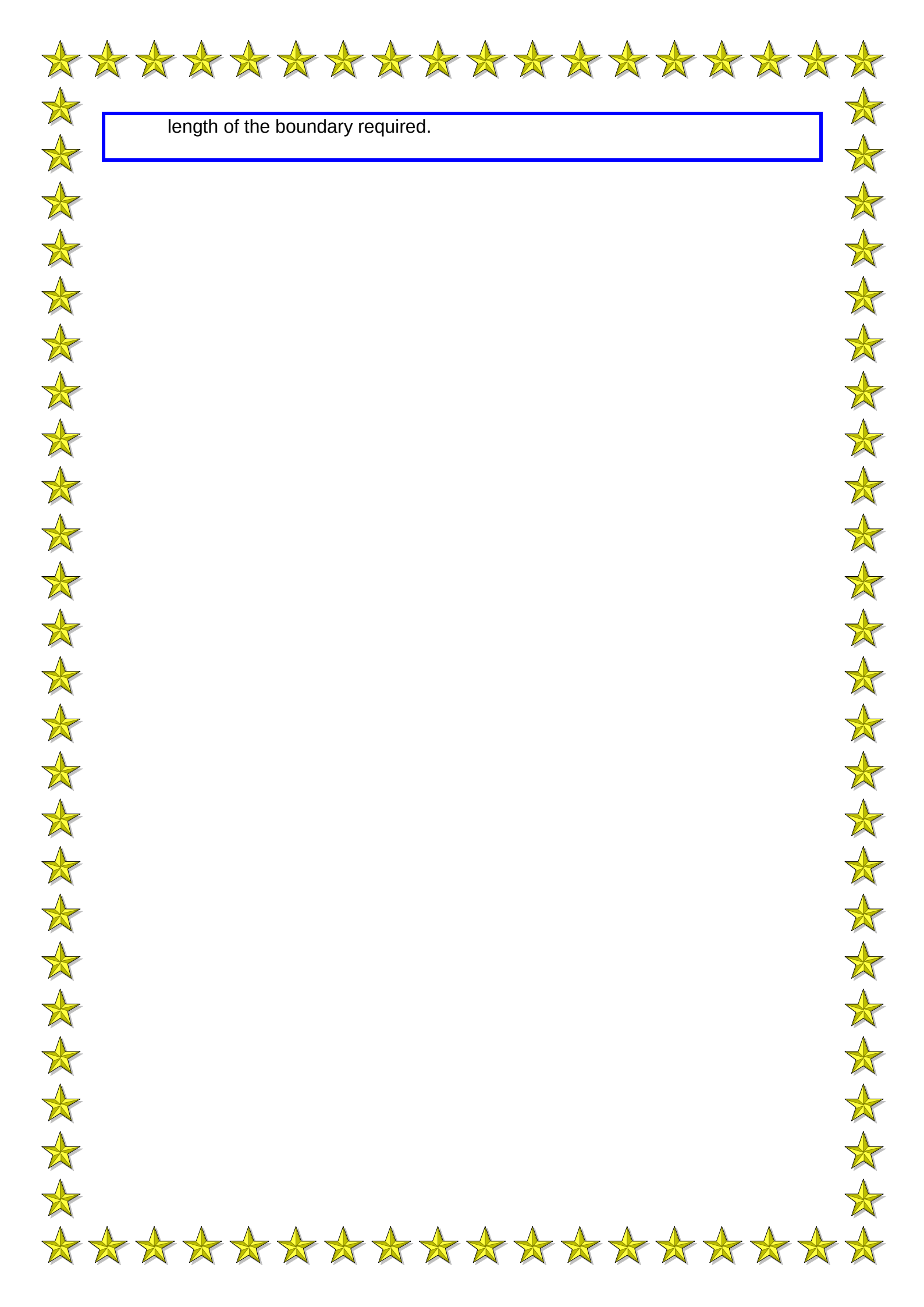
Input	Process	Output
Radius r	Area = $\pi \times r^2$	Area of circle

Why PAC is Important

- Helps **understand the problem clearly**
- Makes **algorithm writing easier**
- Avoids confusion before coding
- Acts as a **bridge between problem and program**



1. A shopkeeper receives three different amounts of money from three customers. Write a program to find the total amount received from all three customers.
2. A teacher gives a number to a student and asks the student to find half of that number. Write a program to calculate and display the result.
3. A school wants to put a boundary around a rectangular playground. Given the length and width of the playground, write a program to calculate the total



length of the boundary required.